

Building OWL Ontologies

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1 Project Introduction

Ontology refers to a structured framework that represents knowledge as a set of concepts within a domain, and the relationships between those concepts. It serves as a formal naming and definition of the categories, properties, and relations among the entities in a specific area of interest.

When it comes to ontology engineering, it is defined as the systematic process of designing, building, and maintaining formal representations of knowledge that entail concepts, relationships, and rules within specific domains.

The main aim of this project relies on the specialization in the domain of traffic regulations. This report contains the full process and completion of the tasks distributed across the project development timestamps. From the first contact with Protégé to the creation of a formal knowledge model. Each task includes the prompt and its own set of competency questions, which were executed in Chat-GPT 5.5. thinking model.

The whole process is divided into two stages, the Road-Sign Building Stage, where tasks from 1 to 4 were oriented towards building the taxonomy and incrementally improving it, and the Building-Stage Assessment, where tasks from 5 to 6 focused on the application validation.

1.1 Prompt Template

The ontology was carried out using LLM-based ontology engineering prompts. This subsection includes the prompt template used for each, to ensure an equal structure in order to make a better analysis of the outputs obtained at each phase.

Prompt for Ontology Generation

Role: You are an expert Ontology Engineer specializing in the domain of traffic regulations.

Objective: [describe the task to do].

Scope and Domain:

- Domain: [define the concrete scope of the domain to be modeled].
- Key Entities: [provide key entities that need to be included in the ontology].
- Boundaries: [define what NOT to include to prevent scope creep].

Competency Questions: The ontology must be able to answer the following questions:

- Competency Question 1.
- Competency Question 2.
- ...

Structural Requirements: [describe particular implementation requirements for this task].

Output Format: [specify the concrete output format (XML/RDF, Turtle, etc)].

This general structure provided a controlled modeling stage plus guaranteed the prevention of unnecessary ontology redesigning.

The final building-stage ontology entails the following components:

- Class hierarchy rooted in **RoadSign**.
- Vienna Convention road-sign categories.

- MTDS dataset classes aligned with the ontology.
- `owl:subClassOf` hierarchy axioms.
- `owl:equivalentClass` mappings.
- Object properties with domains and ranges.
- Data properties with domains and ranges.
- Property characteristics.
- `rdfs:comment` definitions for the main road-sign hierarchy.

1.2 Competency Questions (CQs)

1.2.1 CQs about the taxonomy of road signs

- What are the main categories of road signs in the Vienna Convention?
- Is a danger warning sign a type of road sign?
- Is a prohibitory sign a type of regulatory sign?
- Is a mandatory sign a type of regulatory sign?
- Which classes are subclasses of `RoadSign`?
- Which signs belong to the category of additional panels?
- Which signs are informative rather than regulatory?

1.2.2 CQs about road sign properties

- What shapes are associated with each traffic sign category?
- What is the numerical speed limit value represented by a speed limit sign?
- What is the minimum speed value represented by a compulsory minimum speed sign?
- What is the advisory speed value represented by an advisory speed sign?
- What is the maximum permitted height represented by a height restriction sign?
- What is the maximum permitted width represented by a width restriction sign?
- What is the maximum permitted length represented by a length restriction sign?
- What is the maximum permitted laden mass represented by a weight restriction sign?
- What is the maximum permitted axle mass represented by an axle-weight restriction sign?
- What distance value is displayed on an additional panel?
- What textual information is displayed on an information sign?
- What destination text is displayed on a direction sign?
- What service name is displayed on a service sign?
- What road-sign code is associated with a sign class?
- Which properties are functional?
- Which object properties are inverse properties of each other?

1.2.3 CQs about the pragmatics of road signs

- Which type of road user does a sign apply to?
- Which type of vehicle does a regulatory sign apply to?
- What regulation is expressed by a regulatory sign?
- What prohibition is expressed by a prohibitory sign?
- What obligation is expressed by a mandatory sign?
- What restriction is expressed by a restriction sign?
- What hazard is indicated by a warning sign?
- What road condition is indicated by a warning sign?
- What priority rule is expressed by a priority sign?
- What direction is indicated by a direction sign?
- What destination is indicated by a direction sign?
- What service is indicated by a service sign?
- Which additional panel modifies a main road sign?
- Which main sign is modified by an additional panel?
- Can the ontology distinguish between signs expressing prohibitions, obligations, restrictions, hazards, directions, services, and priority rules?

1.2.4 CQs relevant to the particular dataset

- Which MTDS classes are represented in the ontology?
- Which MTDS classes are subclasses of existing road-sign categories?
- Which MTDS classes are equivalent to Vienna Convention road-sign classes?
- Are all MTDS classes connected to the **RoadSign** hierarchy?
- Which MTDS classes belong to warning signs?
- Which MTDS classes belong to regulatory signs?
- Which MTDS classes belong to mandatory signs?
- Which MTDS classes belong to prohibitory signs?
- Which MTDS classes belong to direction or indication signs?
- Are there MTDS classes that do not have an exact Vienna Convention equivalent?
- What MTDS dataset label is associated with a class?
- What image filename contains a given MTDS annotation?
- What are the width and height of an MTDS image?
- What dataset split does an MTDS image belong to?
- What are the bounding-box coordinates of an MTDS annotation?
- Are MTDS metadata properties clearly distinguished from road-sign semantic properties?

2 Road-Sign Ontology Building Stage

To complete the building stage of the road-sign ontology, four main branches characterized the workflow of this stage. The ontology started as a formal taxonomy of Vienna Convention road signs and was progressively extended with MTDS dataset alignment, object and data properties and finally with property characteristics and class definitions.

2.1 Creation of the Taxonomy - Vienna Convention Road-Sign

The first step carried out was based on creating the initial road-sign ontology as a formal class hierarchy. As suggested in the prompt, it was only needed to satisfy the Part I, Annex 1 of the Vienna Convention in the main taxonomy produced.

Prompt for Task 1

Role: You are an expert Ontology Engineer specializing in the domain of traffic regulations.

Objective: Develop a formal taxonomy of road signs as an IS-A class hierarchy, written in OWL2, as the first step of an ontology engineering process.

Scope and Domain:

- Domain: The taxonomy should cover all road signs and their taxonomic structure, as described in Part I, Annex 1, of the attached PDF document "*Conv_road_signs_2006v_EN.pdf*".
- Key Entities: The class hierarchy should be rooted in a class named **RoadSign**. Subclasses should follow the naming conventions in the attached PDF document.
- Boundaries: Only declare **RoadSign** and the subclasses mentioned in Part I, Annex 1, of the attached PDF document, but no other ontological entities. Do not generate the whole ontology in this step.

Competency Questions: When developing the taxonomy, take into account that the final ontology developed should be able to answer the following questions, CQs about taxonomy.

- What are the main categories of road signs in the Vienna Convention?
- Is a danger warning sign a type of road sign?
- Is a prohibitory sign a type of regulatory sign?
- Is a mandatory sign a type of regulatory sign?
- Which classes are subclasses of **RoadSign** ?
- Which signs belong to the category of additional panels?
- Which signs are informative rather than regulatory?

Structural Requirements: Add disjointness axioms to keep classes and subclasses separate whenever this is the case.

Output Format: Generate the class hierarchy in RDF/XML syntax.

2.1.1 Output Analysis

The ontology obtained contained the following:

- A formal road-sign taxonomy.
- Main Vienna Convention categories.
- Subclass relations using `rdfs:subClassOf`.
- No object and data properties.
- No MTDS classes and no individuals.

After revising the generated ontology, the general structure seemed correct. The ontology used **RoadSign** as the root class and organized the main categories according to the Vienna Convention. The Convention distinguished three main kinds of road signs: danger warning signs, regulatory signs, and informative signs. Regulatory signs were then divided into priority signs, prohibitory or restrictive signs, mandatory signs, and special regulation signs. Informative signs were divided into information, facilities or service signs, direction, position or indication signs, and additional panels.

The use of **owl:AllDisjointClasses** was also appropriate in this step. Disjointness axioms helped ensure that different categories were not accidentally classified as the same type of sign. For example, a danger warning sign should not also be classified as a regulatory sign, and a mandatory sign should be kept separate from a prohibitory or restrictive sign when the Convention treated them as different categories.

First, **AdditionalPanel** was placed under **InformativeSign**, which followed the structure of Article 5 of the Convention. However, from an ontological point of view, additional panels usually function as auxiliary elements that modify another sign rather than as independent road signs. Thought this should be taken into account when properties were added.

There's also some class names that were too broad or ambiguous such as **LaneRegulationOrDangerWarningSign**. Multiple trials were needed to refine the actual taxonomy, it was done through reviewing and feedback to the LLM in order to modify it and minor detailed fixing, which was done by hand, basically the naming of classes and hierarchy positioning.

2.1.2 Assessment

The Task 1 output was correct and useful as a foundation as it managed to establish the base hierarchy needed for all later extensions developed in the next tasks. The main strength of this stage was that it kept the ontology simple and focused, which makes sense as the only goal was to build the basic taxonomy. **RoadSign** was correctly treated as the root class, and all main road-sign categories were placed below it. The output was intentionally limited and did not yet include semantic relationships or dataset alignment.

2.2 Alignment with MTDS Dataset

This second task consisted of aligning the obtained road-signed taxonomy with MTDS dataset classes. To do so, we added the missing subclasses in addition to declaring equivalence mappings where an existing ontology class had the same meaning as an MTDS class.

Prompt for Task 2

Role: You are an expert Ontology Engineer specializing in the domain of traffic regulations and traffic sign classification.

Objective: Adjust the existing road-sign ontology concept hierarchy so that it is aligned with the classes used in the MTDS dataset. The task is to compare the current ontology classes with the MTDS dataset classes, identify missing concepts, add the necessary subclass relations using `owl:subClassOf`, and declare semantic equivalences between ontology classes and MTDS classes using `owl:equivalentClass` where appropriate.

Scope and Domain:

- Domain: The ontology models road traffic signs (based on Part I, Annex 1, of the attached PDF document "Conv_road_signs_2006v_EN.pdf") and their classification according to traffic regulation categories, with a specific focus on aligning the ontology hierarchy with the class structure of the MTDS traffic-sign dataset.
- Key Entities: The ontology must include and organise the following types of entities, **RoadSign** as root class, **RegulatorySign**, **WarningSign**, **ProhibitorySign**, **MandatorySign**, **PrioritySign**, **InformationSign**, **DirectionSign**, **ServiceSign**, **AdditionalPanel**, **SpecialRegulationSign**, MTDS dataset sign classes and any missing subclasses required to represent the MTDS taxonomy correctly.
- Boundaries: This task is restricted to redesigning the ontology from scratch, changing **RoadSign** as the root class, replacing the Vienna Convention hierarchy and removing any existing classes or subclass axioms.

Competency Questions:

- Which MTDS classes are represented in the ontology?
- Which MTDS classes are subclasses of existing road-sign categories?
- Which MTDS classes are equivalent to Vienna Convention road-sign classes?
- Are all MTDS classes connected to the **RoadSign** hierarchy?
- Which MTDS classes belong to warning signs?
- Which MTDS classes belong to regulatory signs?
- Which MTDS classes belong to mandatory signs?
- Which MTDS classes belong to prohibitory signs?
- Which MTDS classes belong to direction or indication signs?
- Are there MTDS classes that do not have an exact Vienna Convention equivalent?

Structural Requirements:

- Preserve the existing **RoadSign** taxonomy.
- Add new MTDS classes.
- Connect MTDS classes to the existing hierarchy.
- Use `owl:equivalentClass` only for exact semantic matches.
- Do not use `owl:equivalentClass` for broader, narrower, or merely related concepts.
- Ensure that every MTDS class is ultimately connected to **RoadSign**.

Output Format: Provide the output in RDF/XML OWL format. Ready to download.

2.2.1 Output Analysis

Now the ontology consisted of:

- The original Vienna Convention hierarchy.
- MTDS dataset classes.
- Additional subclass axioms.
- `owl:equivalentClass` mappings.
- No object nor data properties.

We basically achieved an extended taxonomy by the addition of the classes and mappings.

The overall structure followed the requirements of the prompt reasonably well. The ontology used `RoadSign` as the root and incorporates both the Vienna Convention hierarchy and the MTDS dataset taxonomy. More than four hundred MTDS classes had been added and connected to the ontology through subclass relations, ensuring that dataset labels were linked to the hierarchy rather than remaining as isolated classes.

The model preserved the main road-sign categories required by the prompt, including `RegulatorySign`, `WarningSign`, `PrioritySign`, `MandatorySign`, `ProhibitorySign`, `SpecialRegulationSign`, and the informative-sign branch. In addition MTDS classes such as `MTDSWarningSignClass`, `MTDSMandatoryRegulatoryClass`, `MTDSDirectionIndicationClass` and related grouping classes were introduced to organize the large number of dataset categories. This allowed for the competency questions to be answered more easily.

A positive outcome was the use of `owl` for a number of clear semantic correspondences between Vienna Convention concepts and MTDS classes. Examples include alignments such as `GiveWaySign` with the MTDS yield sign class, `HospitalSign` with the MTDS hospital sign class, and `MotorwaySign` with the corresponding MTDS class. These equivalence axioms enable reasoning systems to infer that instances of either class belong to the same conceptual category.

The generated ontology also shows awareness that not every MTDS class has an exact Vienna Convention equivalent. In several cases, MTDS classes were attached through `subclassOf` relations rather than `equivalentClass`. This is an appropriate modeling choice because many MTDS labels represent visual variants and dataset-specific categories that do not perfectly correspond to a Convention sign.

However, some modeling issues remained as the ontology contained both `WarningSign` and `DangerWarningSign`, which represent the same conceptual category. Which might introduce redundancy and ambiguity in the hierarchy. A clearer design would select a single preferred concept.

Another issue was the introduction of dataset-oriented classes such as `MTDSDatasetSignClass` and `MTDSOtherSignClass`. These classes did not originate from the Vienna Convention and could blur the distinction between conceptual road-sign categories and dataset management categories.

The ontology also includes comments explaining alignment decisions and corrections. These annotations improve transparency and achieve an easier understanding why certain MTDS classes were connected through subclass relations rather than equivalence axioms.

2.2.2 Assessment

Overall, the ontology represented a significant improvement over the ontology generated in Task 1. It successfully integrated the MTDS taxonomy, established links between MTDS and Vienna Convention concepts, and ensures that MTDS classes are connected to the `RoadSign` hierarchy. The main weaknesses consisted of minor structural redundancies and the presence of some dataset-specific organizational classes whose ontological status could be refined in future versions.

Therefore we can conclude that it preserved properly the original ontology while including the MTDS alignment. Still, this created the basis for later questions such as:

- Which dataset classes correspond to the Vienna Convention signs?
- Which classes have the exact semantic equivalence?
- Which MTDS categories are subclasses of broader road-sign classes?

2.3 Implementing Object and Data Properties

This task extended the ontology beyond a pure class hierarchy by adding object properties and data properties. Object properties were used to represent relationships between road-sign classes and conceptual entities, while data properties were used to represent literal values such as speed values, height limits, textual labels, image metadata, and annotation metadata.

Visual road-sign properties were also represented in this stage, because several competency questions required querying visual characteristics, such as the shapes associated with traffic sign categories.

Prompt for Task 3

Role: You are an expert Ontology Engineer specializing in traffic regulations, road-sign ontologies, OWL object-property modelling, OWL datatype modelling, and traffic-sign dataset metadata.

Objective: Extend the existing road-sign ontology by adding object properties and data properties. The ontology already contains a class hierarchy rooted in `RoadSign`, Vienna Convention road-sign classes, MTDS dataset classes, `owl:subClassOf` hierarchy axioms, and `owl:equivalentClass` mappings. The task must preserve the existing taxonomy and enrich it with semantic properties and literal-value properties.

Scope and Domain:

- Domain: The ontology models road traffic signs according to the Vienna Convention and aligns them with MTDS dataset classes. This task focuses on adding semantic relationships, visual attributes, numerical values, textual values, and MTDS metadata.
- Key Entities: Existing road-sign classes such as `RoadSign`, `DangerWarningSign`, `RegulatorySign`, `ProhibitorySign` etc.
- Boundaries: Do not redesign the class hierarchy, change `RoadSign` as the root class, remove existing subclass or equivalence axioms, create instance-level image annotations, or model machine learning and computer vision workflows.

Competency Questions:

- Which type of road user does a sign apply to?
- Which type of vehicle does a regulatory sign apply to?
- What prohibition is expressed by a prohibitory sign?
- What obligation is expressed by a mandatory sign?
- What hazard is indicated by a warning sign?
- What direction is indicated by a direction sign?
- What service is indicated by a service sign?
- Which additional panel modifies a main road sign?
- What is the numerical speed limit value represented by a speed limit sign?
- What distance value is displayed on an additional panel?
- What MTDS metadata is represented for images and annotations?

Structural Requirements: Add object properties using `owl:ObjectProperty`, including `rdfs:domain`, `rdfs:range`, and `rdfs:comment`. Add data properties using `owl:DatatypeProperty`, including appropriate XML Schema datatypes. Use `rdfs:subPropertyOf` and `owl:inverseOf` where appropriate. Add conservative class restrictions where needed to support reasoning and competency-question answering, especially for road-user categories, semantic sign functions, numerical values, and visual shape properties.

Output Format: Provide the output in RDF/XML OWL format ready to download

2.3.1 Output Analysis

The output added object properties, data properties, and supporting conceptual classes. Object properties represented semantic relationships such as signs applying to road users or vehicles, signs expressing prohibitions or obligations, signs warning about hazards, signs indicating directions or services, and additional panels modifying main signs.

Visual properties were also included. The object property `hasShape` was used to associate traffic sign categories with shape classes such as `TriangularShape`, `CircularShape`, `RectangularShape`, `OctagonalShape`, `DiamondShape`, and `InvertedTriangularShape`.

2.3.2 Revision Made

During review, some modeling issues were corrected. First, `MTDSImage` and `MTDSAnnotation` were explicitly declared as OWL classes because they were used as domains for metadata data properties. Second, some overly strong sub-property relations for speed and weight values were revised. The broader parent properties `hasSpeedValue` and `hasWeightRestrictionValue` were introduced to avoid incorrect inferences between maximum speed, minimum speed, advisory speed, laden mass, and axle mass properties.

2.4 Property Characteristics and Class Definitions

Prompt for Task 4

Role: You are an expert Ontology Engineer specializing in traffic regulations, road-sign ontologies, OWL modelling, property characteristics, and class definitions.

Objective: Extend the existing road-sign ontology by adding property characteristics and textual definitions for the main road-sign hierarchy classes. The ontology already contains the class hierarchy, MTDS alignment, object properties, and data properties. Therefore, this task must not redesign the hierarchy or remove existing modelling elements.

Scope and Domain:

- Domain: The ontology models road traffic signs according to the Vienna Convention and aligns them with MTDS dataset classes. This task focuses on semantic enrichment and documentation.
- Key Entities: Main road-sign hierarchy classes.
- Boundaries: Do not redesign the class hierarchy, add new road-sign classes, remove existing subclass axioms, remove existing equivalence mappings, or change domains and ranges unless there is a clear logical error.

Competency Questions:

- Which properties are functional?
- Which object properties are inverse properties of each other?
- Which properties should not be functional because they may have multiple values?
- Can the ontology identify that an additional panel modifies a main road sign?
- Can the ontology distinguish between signs expressing prohibitions, obligations, restrictions, hazards, directions, services, and priority rules?
- Can users understand the intended meaning of each main road-sign class through definitions?
- Are MTDS metadata properties clearly distinguished from road-sign semantic properties?

Structural Requirements: Add property characteristics only where logically safe. Add `rdfs:comment` definitions for the main hierarchy classes. Avoid unsafe characteristics on broad semantic object properties.

Output Format: Provide the output in RDF/XML OWL format ready to download.

2.4.1 Output Analysis

The output added property characteristics and textual definitions. Functional-property declarations were added mainly to safe metadata and specific-value data properties. The main road-sign classes also received `rdfs:comment` definitions, making the ontology easier to understand and document.

2.4.2 Revision Made

Some broad object properties and visual/descriptive properties were marked as functional even though they could legitimately have multiple values. These unsafe functional declarations were removed. Functional characteristics were preserved only for safer metadata and specific-value data properties.

2.4.3 Assessment

The Task 4 output improved the semantic precision and documentation of the ontology. The final version was more conservative and avoided incorrect reasoning caused by unsafe property characteristics.

3 Assesment of the Ontology

3.1 Checking the ontology with HerMiT

The ontology was checked in Protégé using the semantic reasoner HerMiT. The purpose of this step was to verify the logical consistency of the ontology and inspect the inferred class hierarchy.

The inferred hierarchy was largely the same as the asserted hierarchy. This indicated that the main taxonomy had already been explicitly modeled through `subClassOf` axioms and that the reasoner did not need to infer many additional visible subclass relations.

In the inferred hierarchy, `RoadSign` appeared as a direct subclass of `owl:Thing`, together with several auxiliary conceptual classes such as `RoadUser`, `RoadHazard` etc. This result was expected because these auxiliary classes were not road-sign classes, but rather supporting concepts used as domains or ranges of object and data properties.

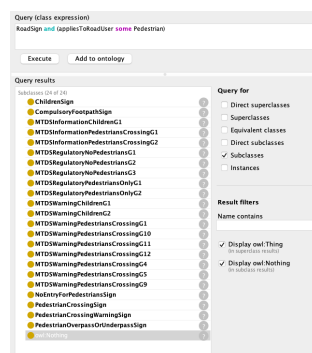
Therefore, the reasoner validation indicated that the ontology was consistent and that the asserted modeling decisions were compatible with OWL reasoning.

3.2 DL queries

After validating the ontology with the semantic reasoner, a set of competency questions were evaluated using the DL Query tab in Protégé. The purpose of these queries was to test whether the ontology could retrieve road-sign classes according to their semantic function, affected road-user category, vehicle category and datatype restrictions.

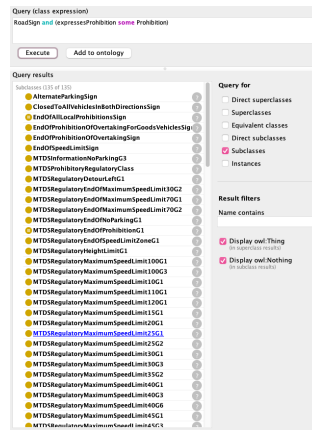
- Which road signs are relevant to pedestrians? **Query:**

`RoadSign and (appliesToRoadUser some Pedestrian)`



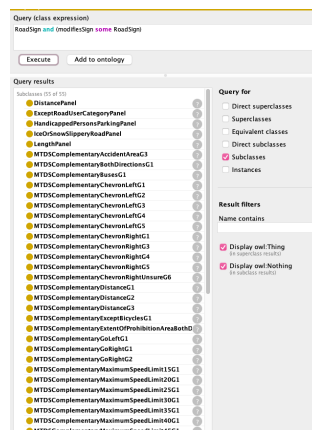
- What prohibition is expressed by a prohibitory sign? **Query:**

RoadSign and (expressesProhibition some Prohibition)



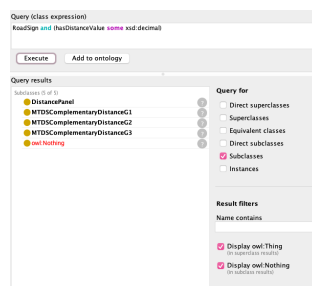
- Which additional panel modifies a main road sign? **Query:**

RoadSign and (modifiesSign some RoadSign)



- What distance value is displayed on an additional panel? **Query:**

RoadSign and (hasDistanceValue some xsd:decimal)



These results demonstrated that the ontology was not only a class hierarchy, but also supported reasoning over semantic restrictions added to road-sign classes.

3.3 SPARQL queries

- Which MTDS classes are equivalent to Vienna Convention road-sign classes?

```
PREFIX owl: <http://www.w3.org/2002/07/owl#>

SELECT ?class ?equivalentClass
WHERE {
    ?class owl:equivalentClass ?equivalentClass .
}
ORDER BY ?class
```

This query retrieved all explicit `owl:equivalentClass` mappings. The result validated the MTDS alignment stage by showing which ontology classes were mapped as exact semantic equivalents.

- Which object properties describe road-sign semantics?

```
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT ?property ?domain ?range
WHERE {
    ?property rdf:type owl:ObjectProperty .
    OPTIONAL { ?property rdfs:domain ?domain . }
    OPTIONAL { ?property rdfs:range ?range . }
}
ORDER BY ?property
```

This query retrieved object properties together with their domains and ranges. It validated that semantic relations such as signs applying to road users, signs expressing regulations, and additional panels modifying main signs were formally represented.

- Which object properties are inverse properties of each other?

```
PREFIX owl: <http://www.w3.org/2002/07/owl#>

SELECT ?property ?inverseProperty
WHERE {
    ?property owl:inverseOf ?inverseProperty .
}
ORDER BY ?property
```

3.4 Python / Owlready2 Query

In addition to DL and SPARQL queries, one competency question was evaluated using a Python script with the Owlready2 library. The aim was to test whether the ontology could be inspected via programming outside Protégé.

- What shapes are associated with each traffic sign category?

This competency question belonged to the road-sign properties category, since the Vienna Convention described several road-sign categories partly through their visual characteristics. In the ontology, this information is represented through the object property `hasShape` and shape classes such as `TriangularShape`, `CircularShape`, `RectangularShape`, `OctagonalShape`, `DiamondShape`, and `InvertedTriangularShape`.

The Owlready2 script loads the ontology, identifies the `hasShape` object property, iterates through the main traffic-sign categories, checks each category and its subclasses for restrictions using `hasShape`, and prints the shape fillers found for each category.

The query returned shape information for the traffic-sign categories that had explicit `hasShape` restrictions. The result showed, for example, that danger warning signs are associated with triangular shapes, prohibitory or restrictive signs and mandatory signs are associated with circular shapes, additional panels are associated with rectangular shapes, STOP signs are associated with octagonal shapes, give-way signs are associated with inverted triangular shapes, and priority road signs are associated with diamond shapes.

Ontology loaded successfully: road_sign_ontology_Sara_Maria_Gussem.owl
Reasoner skipped in Python. Using asserted ontology axioms only.

Shape property used: hasShape

Shapes associated with each traffic sign category:

- DangerWarningSign:
 - TriangularShape
- WarningSign:
 - TriangularShape
- RegulatorySign:
 - CircularShape
 - DiamondShape
 - InvertedTriangularShape
 - OctagonalShape
 - RectangularShape
- PrioritySign:
 - CircularShape
 - DiamondShape
 - InvertedTriangularShape
 - OctagonalShape
 - RectangularShape
- ProhibitorySign:
 - CircularShape
- ProhibitoryOrRestrictiveSign:
 - CircularShape
- MandatorySign:
 - CircularShape
- SpecialRegulationSign:
 - RectangularShape
- InformativeSign:
 - RectangularShape
- InformationSign:
 - RectangularShape
- DirectionSign:
 - RectangularShape
- DirectionPositionOrIndicationSign:
 - RectangularShape
- ServiceSign:
 - RectangularShape
- AdditionalPanel:
 - RectangularShape

This result confirmed that the ontology could properly answer a road-sign property competency question programmatically. It also showed that the visual characteristics added during the object-property enrichment stage could be retrieved using Owlready2. Therefore, the ontology was not only usable through

Protégé, but could also be queried through external Python scripts.

4 Conclusion

This project developed and validated a road-sign ontology through an incremental ontology engineering process. The ontology first represented the Vienna Convention road-sign taxonomy, then aligned it with MTDS dataset classes, and finally enriched it with object properties, data properties, property characteristics, class definitions, and selected reasoning-oriented restrictions.

The validation stage showed that the ontology was consistent under the HermiT reasoner and that it could successfully answer competency questions through DL queries, SPARQL queries, and Python/Owlready2 scripts. DL queries were used to test class-level reasoning over semantic restrictions, SPARQL queries were used to inspect equivalence mappings and property structures, and Owlready2 were used to query visual shape information via programming.

Overall, the final ontology was not only a static class hierarchy, but a richer semantic model capable of representing road-sign categories, dataset alignment, sign meanings, literal attributes, metadata, and selected visual properties.